Facilitative Processing of Gender in the Absence of Overt Agreement Morphology in Heritage Spanish Speakers

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Background. Heritage speakers (HSs) often demonstrate non-target-like production and comprehension of grammatical gender in offline tasks but target-like processing in real-time comprehension **[1]**. Prior work on facilitative processing in HSs has primarily examined gender agreement with overt agreement cues, e.g., Spanish *el* vs. *la*, revealing that HSs are able to use overt gender cues on an article to facilitate the lexical retrieval of a subsequent noun (e.g., **[2]**). However, despite various offline studies showing HSs' divergence from baseline speakers in interpreting null elements **[3]**, it remains unexplored whether HSs exhibit target-like facilitative processing based on null agreement cues. Existing work on facilitative processing of gender **[4]** showed that baseline Russian speakers use both salient feminine cues (bisyllabic on verbs and adjectives) and less salient masculine cues (monosyllabic on verbs and null on adjectives), whereas HSs only used the salient feminine cues, but not the less salient masculine cues. Therefore, **[4]** suggested that morphological saliency of the agreement marker affects HSs' ability to access and integrate gender features during spoken word recognition. Importantly, however, **[4]** could not determine whether this asymmetry derived from feminine gender's morphological overtness or its markedness as a gender feature.

Aims. We examine facilitative processing of grammatical gender on null vs. overt agreement markers on the indefinite article *algún* 'some' in baseline speakers (BSs) and heritage speakers (HSs) of Spanish. In singular, agreement marking is null for masculine and overt for feminine. In plural, agreement is overt for both genders (Table 1). This asymmetry allows for testing whether HSs can use null agreement morphology for predictive processing and to disentangle this effect from the markedness of the gender feature.

Methods. In a visual-world eye-tracking experiment with 16 masculine and 16 feminine target nouns (128 critical trials, 64 fillers), participants (current N= 20 HSs; 2 BSs; planned N= 35 HSs; 30 CSs) hear a question *"¿Hay algún(-Ø/-a/-os/-as) libro(s)/regla(s)?"* 'Is/are there some book(s)/ruler(s)?' and view three images on the screen (a target, a competitor that either matches or mismatches target in gender or number, and a distractor that differs from target in gender and number) (Figure 1). Participants decide whether the item they hear is on the screen, by clicking its image if present or pressing the spacebar if absent. Facilitative processing of gender is expected to be evidenced by higher proportion of looks to target in the **Gender-informative** condition compared to the **Uninformative** condition (all conditions presented in Table 2). Participants also complete a background questionnaire (adapted from LEAP-Q, **[5]**), picture-naming task (MINT-Sprint, **[6]**), and vocabulary recognition task (LexTale, **[7]**).

Predictions. If HSs use null agreement as a predictive cue, we expect facilitative effects for both feminine and masculine targets in singular and plural forms. This would suggest that the activation of abstract gender features leads to facilitative processing of gender agreement cues, regardless of morphological overtness. If HSs do not use null agreement for facilitative processing, we expect facilitative effects for feminine singular and both plural conditions, but not in masculine singular targets. This would suggest morphological overtness of agreement cues facilitates processing of gender. However, if HSs demonstrate facilitative processing in neither the masculine singular or plural, this would suggest that the markedness of gender feature rather than its morphological overtness modulates HSs' gender processing.

Preliminary Findings. Cluster-based permutation analyses were used to identify clusters of 50ms-time bins in which looks to the target differed by condition after cue onset. We compared looks to target across **Gender-informative** and **Uninformative** conditions. Preliminary data from HSs (n=20) show a significant effect of gender in M-SG and M-PL (Figure 2). However, due to the lack of consistent effects across all conditions (Figure 2), we refrain interpreting this pattern as evidence for this is facilitative processing until data collection is completed.

Figure 1. Sample Visual Display for **M-PL** targets (e.g. *las reglas* 'rulers') in **Uninformative** (left) vs. **Gender-informative** (right) conditions. Participants hear *"¿Hay algun**os libros**?"* and see images as in below. The names of conditions and nouns are provided for abstract readers:

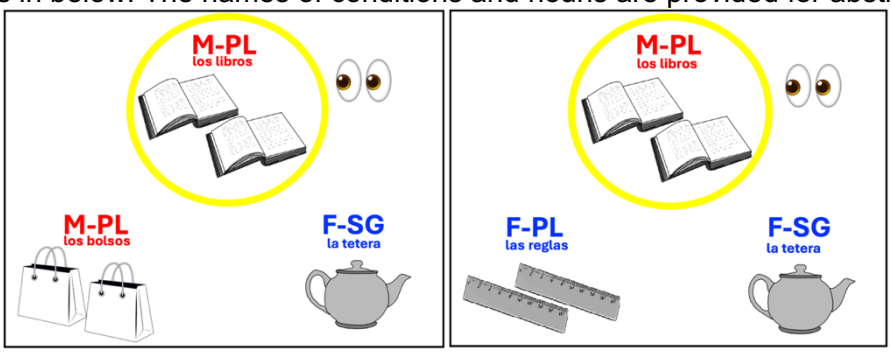


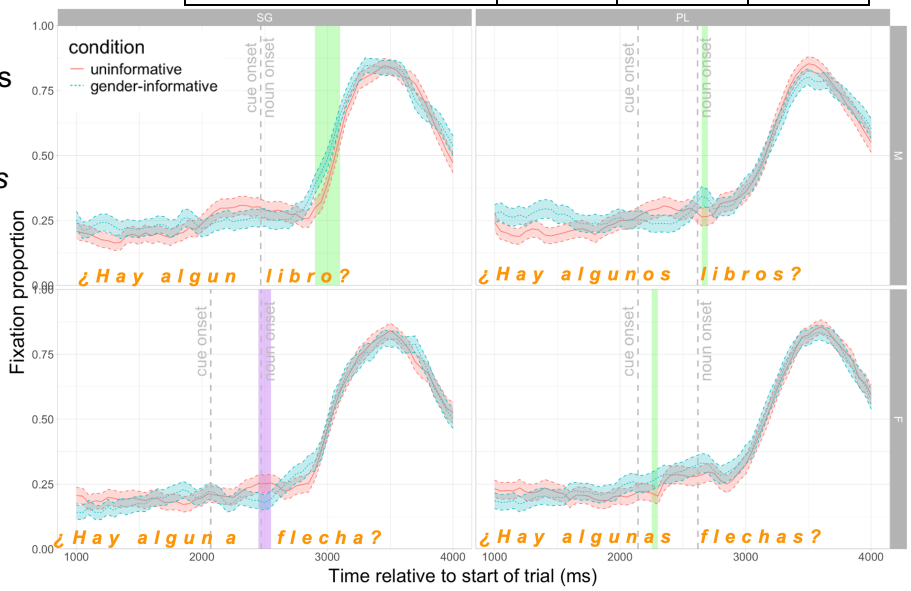
Table 1. Distribution of gender and number agreement marking on the indefinite article *algún*

	Singular	Plural
Masc.	algún.∅	algún-os
Fem.	algun-a	algun-as

Table 2. Experimental Conditions for **M-PL** targets

	Target	Comp.	Dist.
Uninformative	M-PL	M-PL	F-SG
Doubly informative	M-PL	F-SG	F-SG
Number-informative	M-PL	M-SG	F-SG
Gender-informative	M-PL	F-PL	F-SG

Figure 2. Preliminary HS data (n=20): Looks to targets in the gender-informative condition *(sig. clusters for contrast with uninformative condition in the predicted direction marked in green; sig. clusters for contrast with uninformative condition in the opposite direction marked in purple)*



References. [1] Keating, G. D. (2022). The effect of age of onset of bilingualism on gender agreement processing in Spanish as a heritage language. *Language Learning, 72*(4), 1170–1208. **[2]** Fuchs, Z. (2021). Facilitative use of grammatical gender in Heritage Spanish. *Linguistic Approaches to Bilingualism*. **[3]** Sorace, A., & Serratrice, L. (2009). Internal and external interfaces in bilingual language development: Beyond structural overlap. *Int. J of Bil, 13*(2), 195–210. **[4]** Fuchs, Z., & Sekerina, I. (2025). New evidence for the role of morphological markedness of gender agreement cues in monolingual and heritage-bilingual facilitative processing. *Essays in Honor of Maria Polinsky*. **[5]** Marian, V., Blumenfeld, H. K., & Kaushanskaya, M. (2007). The Language Experience and Proficiency Questionnaire (LEAP-Q). *JSLHR, 50*(4), 940–967. **[6]** Garcia, D.L., Gollan, T.H. (2022). The MINT Sprint: Exploring a Fast Administration Procedure with an Expanded Multilingual Naming Test. *J Int Neuropsychol Soc*. **[7]** Lemhöfer, K., & Broersma, M. (2012). Introducing LexTALE: a quick and valid Lexical Test for Advanced Learners of English. *Behavioral research methods, 44*(2), 325–343.